

D 2 A

$$f(x) = \sin(x)$$

$$g(x) = x^2$$

$$h(x) = 2 + x$$

$$(h \circ g \circ f)(x) = x \xrightarrow{f} \sin(x) \xrightarrow{g} (\sin(x))^2 \xrightarrow{h} 2 + (\sin(x))^2$$

D 2 B

$$f(x) = \cos(x)$$

$$g(x) = x^2$$

$$h(x) = 3 - x$$

$$(h \circ g \circ f)(x) = x \xrightarrow{f} \cos(x) \xrightarrow{g} (\cos(x))^2 \xrightarrow{h} 3 - (\cos(x))^2$$

D 7 A

$$f(x) = e^{\sin(x)} \quad f(0) = e^{\sin(0)} = e^0 = 1$$

$$f'(x) = e^{\sin(x)} \cdot \cos(x) = e^{\sin(x)} \cos(x) \quad f'(0) = e^{\sin(0)} \cdot \cos(0) = e^0 \cdot 1 = 1 \cdot 1 = 1$$

$$f''(x) = (e^{\sin(x)} \cos(x)) (\cos(x)) + (e^{\sin(x)}) (-\sin(x))$$

$$e^{\sin(x)} (\cos(x))^2 - \sin(x) e^{\sin(x)} \quad f''(0) = e^{\sin(0)} (\cos(0))^2 - \sin(0) e^{\sin(0)} = 1 \cdot 1 - 0 \cdot 1 = 1$$

$$1 + x + \frac{x^2}{2}$$

D 7 B

$$f(x) = e^{\cos(x)} \quad f(0) = e^{\cos(0)} = e$$

$$f'(x) = e^{\cos(x)} \cdot (-\sin(x)) = -\sin(x) e^{\cos(x)} \quad f'(0) = -\sin(0) e^{\cos(0)} = 0 \cdot e = 0$$

$$f''(x) = (e^{\cos(x)} \cdot (-\sin(x))) (-\sin(x)) + (e^{\cos(x)}) (-\cos(x))$$

$$e^{\cos(x)} (-\sin(x))^2 + (-\cos(x)) e^{\cos(x)} \quad f''(0) = e^{\cos(0)} (-\sin(0))^2 + (-\cos(0)) e^{\cos(0)} = 1 \cdot 0 + (-1 \cdot e) = -e$$

$$e - e \frac{x^2}{2}$$

D 8 A

$$\int_0^1 \frac{2}{1+x^2} = [2 \arctan(x)]_0^1 = (2 \arctan(1)) - (2 \arctan(0)) = 2(\arctan(1) - \arctan(0)) = 2(\pi/4 - 0) = 2 \cdot \frac{\pi}{4} = \pi/2$$

$$\int \frac{2}{1+x^2} = 2 \int \frac{1}{1+x^2} = 2 \arctan(x)$$

D 8 B

$$\int_{-1}^0 \frac{4}{1+x^2} = 4 [\arctan(x)]_{-1}^0 = 4(\arctan(0) - \arctan(-1)) = 4(\pi/4) = \pi$$

$$\int \frac{4}{1+x^2} = 4 \int \frac{1}{1+x^2} = 4 \arctan(x)$$

D 3 A

$$a_n = \frac{n^2(0^n + (-n)^n)}{n^2 - 7^n} \sim \frac{(-n)^n}{-7^n}$$

on non ha limite perché $(-n)^n$ cresce costantemente più

D 3 B

$$a_n = \frac{n^2(-10)^n + n^2}{n^2 - 7^n} \sim \frac{n^2}{-7^n} = -\infty$$

D 6 A

$$f(1) = 1$$

$$f'(1) = 2$$

$$g(x) = \ln(1 + f(x))$$

$$g'(x) = \frac{1}{1 + f(x)} \cdot f'(x)$$

$$g'(1) = \frac{1}{1 + 1} \cdot 2 = \frac{1}{2} \cdot 2 = 1$$

D 6 B

$$f(1) = 3$$

$$f'(1) = 2$$

$$g(x) = 2 \sqrt{1 + f(x)}$$

$$g' = 2 \cdot \frac{1}{2 \sqrt{1 + f(x)}} \cdot f'(x) = \frac{1}{\sqrt{1 + f(x)}} \cdot f'(x)$$

$$g'(1) = \frac{1}{\sqrt{1 + 3}} \cdot 2 = \frac{1}{\sqrt{4}} \cdot 2 = \frac{1}{2} \cdot 2 = 1$$

5 A

$$f(x) = \begin{cases} x + 2e & x \leq -2 \\ \ln(x^2 + x + 2) & x > -2 \end{cases}$$

$$\lim_{x \rightarrow -2^-} x + 2e = -2 + 2e = -2 + 2e$$

$$f(-2) = -2 + 2e$$

$$\lim_{x \rightarrow -2^+} \ln(x^2 + x + 2) = \ln(4)$$

$$-2 + 2e = \ln(4)$$

$$2e = \ln(4) + 2$$

$$e = \frac{\ln(4) + 2}{2}$$

$$e = \frac{2\ln(2) + 2}{2} = \frac{2\ln(2)}{2} + 1 = \ln(2) + 1$$

5 B

$$f(x) = \begin{cases} x - 2 & x \leq 1 \\ e \ln(x^2 + 2x + 3) & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} x - 2 = 1 - 2 = -1$$

$$f(1) = x - 2 = 1 - 2 = -1$$

$$\lim_{x \rightarrow 1^+} e \ln(x^2 + 2x + 3) = e \ln(6)$$

$$-1 = e \ln(6)$$

$$e \ln(6) = -1$$

$$e = \frac{-1}{\ln(6)}$$

E 2A

1)

$$y = \ln\left(\frac{x}{16} + \frac{1}{x}\right) = \ln\left(\frac{x^2 + 16}{16x}\right)$$

DOMINIO

$$\frac{x^2 + 16}{16x} > 0 \quad \left\{ \begin{array}{l} x^2 + 16 > 0 \\ 16x > 0 \end{array} \right\} \quad \begin{array}{l} \forall x \\ x > 0 \end{array} \quad x \in (0, +\infty)$$

SEGNO

$$\ln\left(\frac{x^2 + 16}{16x}\right) > 0 \Rightarrow \frac{x^2 + 16}{16x} > e^0 \Rightarrow \frac{x^2 + 16}{16x} > 1 \Rightarrow \frac{x^2 + 16}{16x} - 1 > 0 \Rightarrow \frac{x^2 + 16 - 16x}{16x} > 0 \quad \left\{ \begin{array}{l} x^2 - 16x + 16 > 0 \\ x > 0 \end{array} \right\} \quad \begin{array}{l} x < 8 - 4\sqrt{3} \vee 8 + 4\sqrt{3} \\ x > 0 \end{array}$$

$$(-\infty, 8 - 4\sqrt{3}) \cup (8 + 4\sqrt{3}, +\infty)$$

LIMITI

$$\lim_{x \rightarrow 0^+} \ln\left(\frac{x^2 + 16}{16x}\right) = \frac{16}{0^+} = +\infty \quad \text{oss. verticale}$$

$$\lim_{x \rightarrow +\infty} \ln\left(\frac{x^2 + 16}{16x}\right) = \ln(+\infty) = +\infty \quad \text{oss. obliquo?}$$

$$\lim_{x \rightarrow +\infty} \ln\left(\frac{x^2 + 16}{16x}\right) / x = +\infty / \infty \quad \text{HOPITAL}$$

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 16}{x^3 + 16x} = 0 \quad \begin{array}{l} \text{semplice} \\ \text{infiniti} \end{array}, \quad \text{no oss. obliquo}$$

MONOTONIA

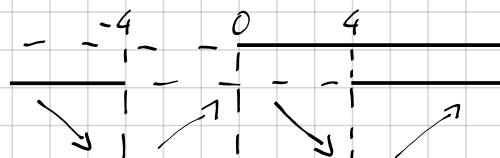
$$y = \ln \left(\frac{x^2 + 16}{16x} \right)$$

$$y' = \frac{x^2 - 16}{x^3 + 16x}$$

$$\frac{x^2 - 16}{x^3 + 16x} > 0 \begin{cases} x^2 - 16 > 0 \\ x^3 + 16x > 0 \end{cases} \begin{cases} x^2 > 16 \\ x(x^2 + 16) > 0 \end{cases} \begin{cases} x < -4 \vee x > 4 \\ x > 0 \\ x^2 + 16 > 0 \end{cases} \begin{cases} x < -4 \vee x > 4 \\ x > 0 \\ \forall x \end{cases}$$

$f(x)$ è crescente per $(4, +\infty) \cup (-4, 0)$

$f(x)$ è decrescente per $(-\infty, -4) \cup (0, 4)$



MASSIMI E MINIMI

$$\frac{x^2 - 16}{x^3 + 16x} = 0$$

$$x^3 + 16x$$

$$x^2 - 16 = 0$$

$$x^2 = 16$$

$$x^2 = \pm 4$$

$$f(-4) = \ln \left(\frac{(-4)^2 + 16}{16(-4)} \right) = \ln \left(\frac{16 + 16}{-64} \right) = \ln \left(\frac{32}{-64} \right) = \ln \left(-\frac{1}{2} \right) \text{ imp}$$

$$f(4) = \ln \left(\frac{4^2 + 16}{16 \cdot 4} \right) = \ln \left(\frac{32}{64} \right) = \ln \left(\frac{1}{2} \right) = -\ln(2)$$

$f(4)$ minimo assoluto

3)

Dal punto (1) sappiamo che $f(x)$ cresce all'infinito e non ha minimo, inoltre $f(x)$ è inferiormente limitata e ha minimo $= \ln(1/2)$ che è anche l'estremo inferiore

$$\sup = +\infty$$

$$\inf = \min = \ln(1/2)$$

2)

$$\ln \left(\frac{x^2 + 16}{16x} \right) = k$$

$$\frac{x^2 + 16}{16x} = e^k$$

$$x^2 + 16 = 16x e^k$$

$$x^2 - 16x e^k + 16 = 0$$

$$\Delta = 256 e^{2k} - 4(1)(16) = 256 e^{2k} - 64$$

$$256 e^{2K} - 64 > 0 \Rightarrow 256 e^{2K} > 64 \Rightarrow e^{2K} > \frac{64}{256} \Rightarrow e^{2K} > \frac{1}{4} \Rightarrow 2K > \ln(1/4) \Rightarrow K > \frac{-\ln(4)}{2} \Rightarrow K > \frac{-2 \ln(2)}{2}$$

$$K > \frac{-2 \ln(2)}{2} \Rightarrow K > -\ln(2)$$

Se $K > -\ln(2)$, l'equazione $f(x) = K$ ha 2 soluzioni

Se $K = -\ln(2)$, l'equazione $f(x) = K$ ha 1 soluzione

Per $K < -\ln(2)$ non ci sono soluzioni perché $-\ln(2)$ è il valore minimo assunto dalla funzione

Eg 3A

1)

Lie on

- 1) una funzione decrescente
- 2) una funzione a termini positivi
- 3) convergente

allora $(-1)^n \sum a_n$ converge

2)

$$y = \sqrt{x} - \ln(x)$$

$$y' = \frac{1}{2\sqrt{x}} - \frac{1}{x} = \frac{x - 2\sqrt{x}}{2x}$$

$$\frac{x-2}{2x} > 0 \quad \left\{ \begin{array}{l} \sqrt{x} - 2 \geq 0 \\ 2x > 0 \end{array} \right\} \left\{ \begin{array}{l} \sqrt{x} \geq 2 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} x \geq 4 \\ x > 0 \end{array} \right\} \quad x \text{ è crescente su } [4, +\infty)$$

3)

$$a_n = \frac{(-1)^n}{\sqrt{2n} - \ln n}$$

sfrutto il criterio di Leibniz

- 1) a_n decrescente

essendo che $\sqrt{2n} - \ln n$ è crescente $1/(\sqrt{2n} - \ln n)$ è decrescente

- 2) a_n è a termini positivi

essendo a_n è il rapporto tra 2 funzioni a termini positivi ovvero a_n è a termini positivi

- 3) a_n converge

$$\lim_{n \rightarrow +\infty} \frac{1}{\sqrt{n} - \ln n} \sim \frac{1}{\sqrt{n}} = 0 \quad \text{per il criterio}$$

8.1A

1)

Siano $f(x)$ e $g(x)$ due funzioni in un intorno x_0 con $g'(x) \neq 0$ e sia $\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0$

se $\lim_{x \rightarrow x_0} f'/g' = l$ allora $\lim_{x \rightarrow x_0} f/g = l$

2)

$$\lim_{x \rightarrow 1} \frac{\sqrt{2-x} - x}{\ln(1)} = \frac{1-1}{0} = \frac{0}{0} = \frac{-\frac{1}{2} - 1}{2} = \frac{\frac{-1-2}{2}}{2} = \frac{-3/2}{2} = -\frac{3}{2} \cdot \frac{1}{2} = -\frac{3}{4}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{2-x} - x}{\ln(1)} = -\frac{3}{4}$$
$$\sqrt{2-x} - x = \frac{1}{2\sqrt{2-x}}, \quad (-1) - 1 = -\frac{1}{2\sqrt{2-x}} - 1$$

$$\ln(x^2) = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$$

3)

$$\int_{\sqrt{2}}^{\sqrt{2}} x(\sqrt{2-x^2} - x^2) = \left[-\frac{1}{3} (2-x^2)^{3/2} - x^4/4 + C \right]_{\sqrt{2}}^{\sqrt{2}}$$

$$\int x \sqrt{2-x^2} - x^3 = \int x \sqrt{2-x^2} - \frac{x^4}{4} = -\frac{1}{3} (2-x^2)^{3/2} - x^4/4 + C =$$

$$\int x \sqrt{2-x^2} = \int \frac{x \sqrt{2-x^2}}{-2x} = \int \frac{\sqrt{y}}{-2} = -\frac{1}{2} \int \sqrt{y} = -\frac{1}{2} \left(\frac{y^{3/2}}{3/2} \right) = -\frac{1}{2} y \cdot \frac{2}{3} = -\frac{2}{6} y^{3/2} = -\frac{1}{3} y^{3/2} = -\frac{1}{3} (2-x^2)^{3/2}$$

$$y = 2 - x^2$$
$$dy = -2x dx$$
$$-\frac{dy}{2x} = dx$$